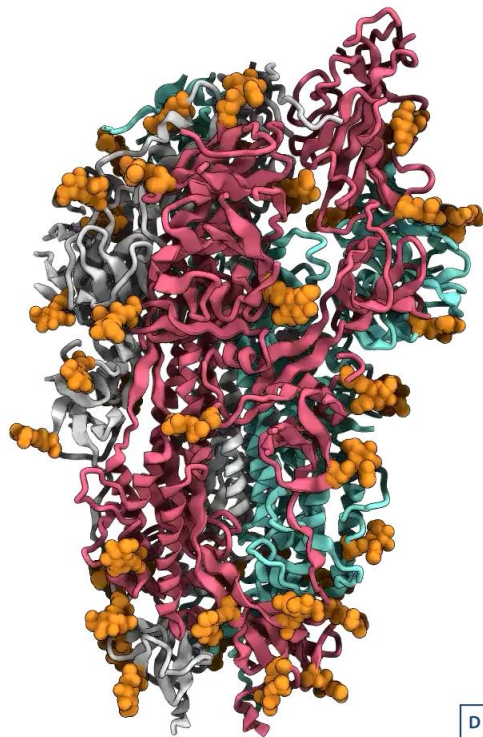


Module 5:

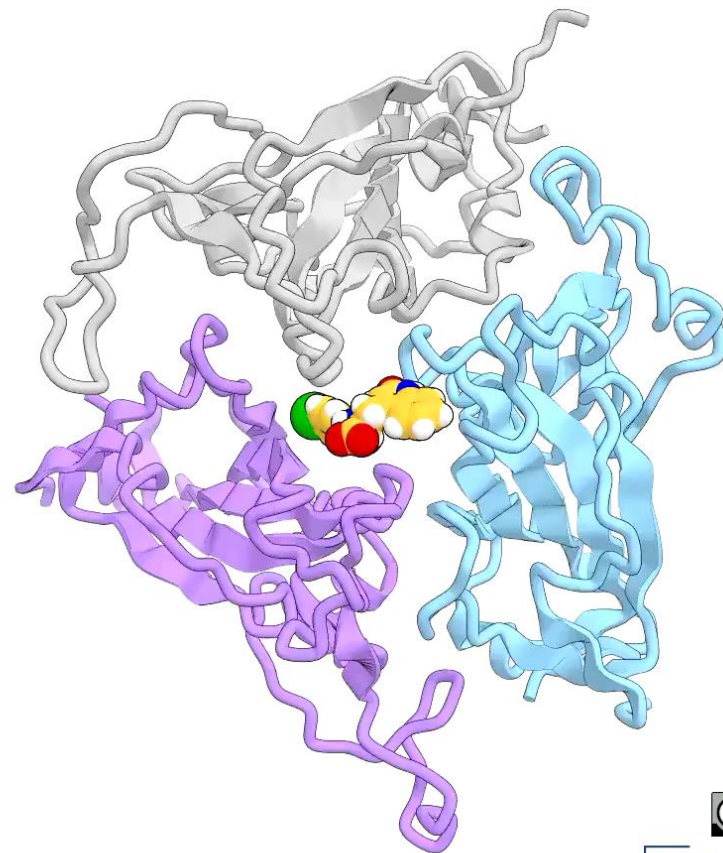
Molecular Dynamics Simulations

0.0 μ s

0.0 μ s



D E Shaw Research



D E Shaw Research

MD simulation trajectories of the trimeric SARS-CoV-2 spike protein and FDA approved or investigational drug molecules binding to a construct of the SARS-CoV-2 trimeric spike protein.

Why Not Quantum Mechanics?

- Modeling the motion of a complex molecule by solving the wave functions of the various subatomic particles would be accurate...

$$\frac{-\hbar^2}{2m} \nabla^2 \Psi + U(x, y, z) \Psi(x, y, z) = E \Psi(x, y, z)$$

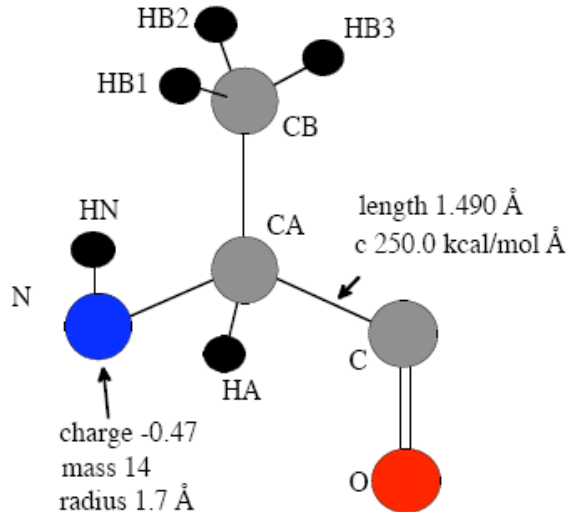
- But it would also be **very** hard to program and take more computing power than anyone has!

Classical or Molecular Mechanics

- Instead of using Quantum mechanics, we can use classical Newtonian mechanics to model our system.
- Calculate the structure and energy of molecules based on nuclear motions.
 - electrons are not considered explicitly
 - will find optimum distribution once position of nuclei are known
 - Born-Oppenheimer approximation of Schrödinger equation
 - nuclei are heavier and move slower than electrons
 - electrons move fast enough to adjust to any nuclei movement

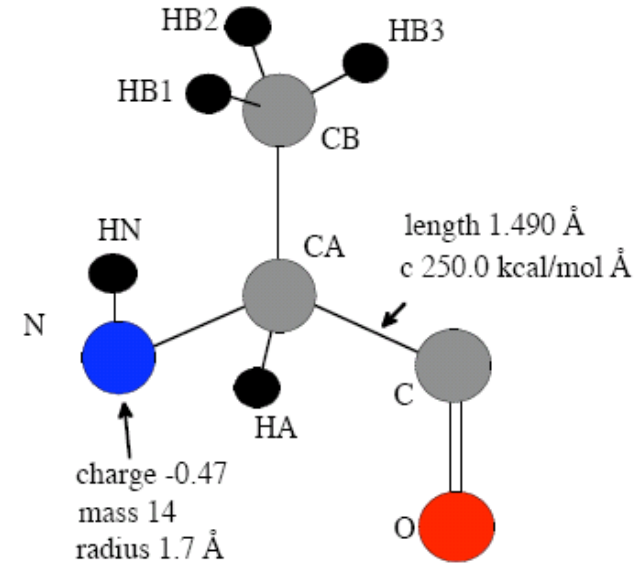
Molecular Mechanics

Molecular mechanics treats a molecule as a collection of weights connected with springs, where the weights represent the nuclei and the springs represent the bonds.



- Collection of **atom types** (to define the atoms in a molecule), **parameters** (for bond lengths, bond angles, etc.) and **equations** (to calculate the energy of a molecule)

MM: Atom Types



In MM, a given element may have several atom types.

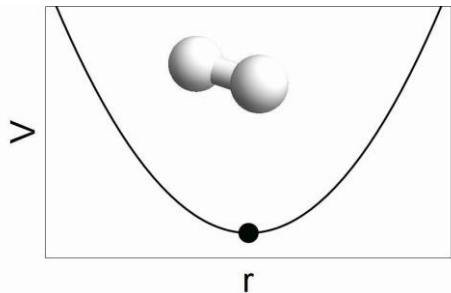
- sp^3 -Hybridized carbons have a tetrahedral bonding geometry
- aromatic carbons have a trigonal bonding geometry.
- C-C bond in the ethyl group differs from a C-C bond in the phenyl ring
- C-C bond between the phenyl ring and the ethyl group differs from all other C-C bonds in ethylbenzene.

Force

Bonded Terms

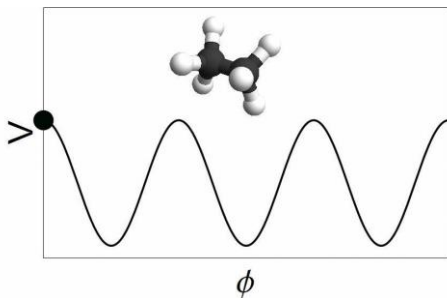
$$V(R) = E_{\text{bonded}} + E_{\text{non-bonded}}$$

Bond Stretch

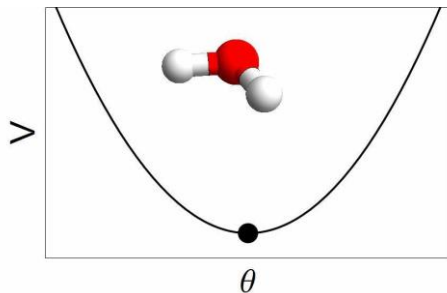


$$E_{\text{str}} = \frac{1}{2} k_{\text{str},ij} (r_{ij} - r_0)^2$$

Dihedral

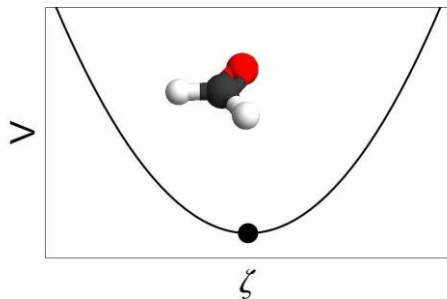


$$E_d = k_{d,ijkl} (1 + \cos(n\Phi - \Phi_s))$$



Bond Angle

$$E_{\text{bend}} = \frac{1}{2} k_{b,ijk} (\theta_{ijk} - \theta_0)^2$$

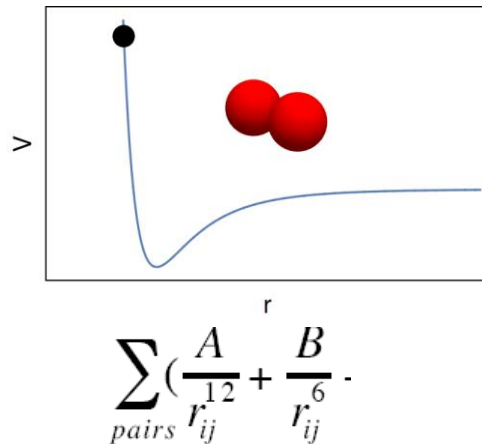


Improper
Dihedral

$$E_{\text{id}} = \frac{1}{2} k_{\text{id},ijkl} (\Phi_{ijkl} - \Phi_0)^2$$

Non-bonded Terms

van der Waals Interaction

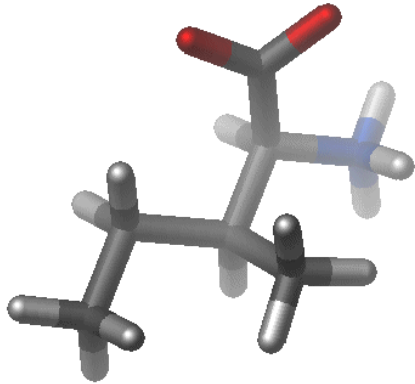


Electrostatic Interaction

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$$

MM: Force Field

Force Field used to calculate the energy and geometry of a molecule.



$$E = \sum_{bonds} k_b (b_0 - b)^2 + \sum_{angles} k_\vartheta (\vartheta_0 - \vartheta)^2 + \sum_{planes} k_\omega (\omega_0 - \omega)^2 + \sum_{dihedrals} k_\varphi (1 - \cos(n\varphi - \delta)) + \sum_{pairs} \left(\frac{A}{r_{ij}^{12}} + \frac{B}{r_{ij}^6} + \frac{Q_i Q_j}{r_{ij}} \right)$$

- Total energy of a molecule is divided into several parts called force potentials, or potential energy equations.
- Force potentials are calculated independently, and summed to give the total energy of the molecule.
- These equations define the potential energy surface of a molecule.

Cars cruise down an expressway at 25 m/s. Engineers are designing an off-ramp in an interchange with a deceleration of -2.0 m/s^2 that lasts 3.0 s.

- What velocity will cars have at the end of the off-ramp?
- What minimum length should the ramp have?

$$v = v_0 + at$$

$$v = (25 \text{ m/s}) + (-2.0 \text{ m/s}^2)(3.0 \text{ s})$$

$$v = 19 \text{ m/s}$$

$$\Delta s = v_0 t + \frac{1}{2}at^2$$

$$\Delta s = (25 \text{ m/s})(3.0 \text{ s}) + \frac{1}{2}(-2.0 \text{ m/s}^2)(3.0 \text{ s})^2$$

$$\Delta s = 66 \text{ m}$$

“Dynamics”

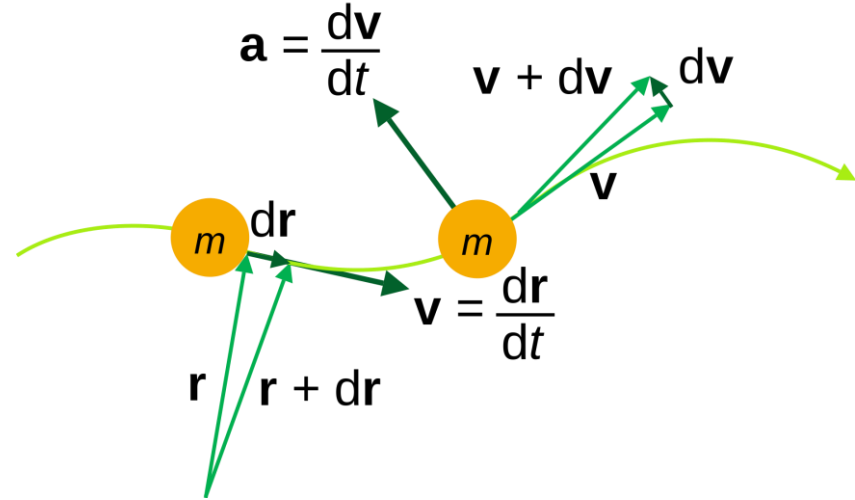
- Newton's second law: $\mathbf{F} = m\mathbf{a}$
 - where $\mathbf{F}$ is force on an atom, m is mass of the atom, and $\mathbf{a}$ is the atom's acceleration
- Recall that: $\mathbf{F}(\mathbf{x}) = -\nabla U(\mathbf{x})$
 - where $\mathbf{x}$ represents coordinates of all atoms, and U is the potential energy function
- Velocity is the derivative of position, and acceleration is the derivative of velocity.
- We can thus write the equations of motion as:

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}$$

$$\frac{d\mathbf{v}}{dt} = \frac{\mathbf{F}(\mathbf{x})}{m}$$

Equation of Motion

- Newton's second law or the equation of motion, $\mathbf{F} = m\mathbf{a}$, where $\mathbf{F}$ is the force exerted on the particle, m is its mass and $\mathbf{a}$ is its acceleration.
- Knowing the force on each atom, one can determine the acceleration of each.
- Integration of the equations of motion then yields a trajectory that describes the positions, velocities and accelerations of the particles as they vary with time.
- From this trajectory, the average values of properties can be determined.



Integration Algorithms

- The potential energy is a function of the atomic positions ($3N$) of all the atoms in the system.
- Due to the complicated nature of this function, there is *no analytical solution* to the equations of motion; they must be solved numerically.
- Numerous **numerical algorithms** have been developed for integrating the equations of motion, such as

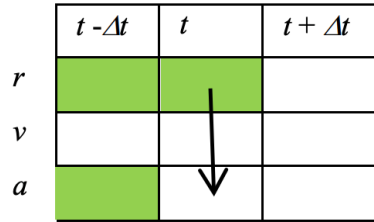
Verlet algorithm

Velocity Verlet

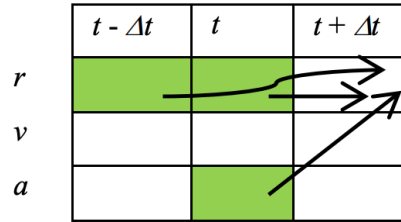
Leap-frog algorithm

(A) The original Verlet method

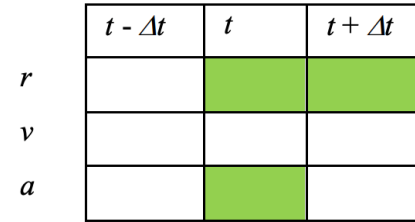
	$t - \Delta t$	t	$t + \Delta t$
r			
v			
a			



	$t - \Delta t$	t	$t + \Delta t$
r			
v			
a			



	$t - \Delta t$	t	$t + \Delta t$
r			
v			
a			



$$r(t + \Delta t) = r(t) + v(t)\Delta t + \frac{f(t)}{2m}\Delta t^2 + \frac{\Delta t^3}{3!} \ddot{r} + \mathcal{O}(\Delta t^4)$$

$$r(t - \Delta t) = r(t) - v(t)\Delta t + \frac{f(t)}{2m}\Delta t^2 - \frac{\Delta t^3}{3!} \ddot{r} + \mathcal{O}(\Delta t^4)$$

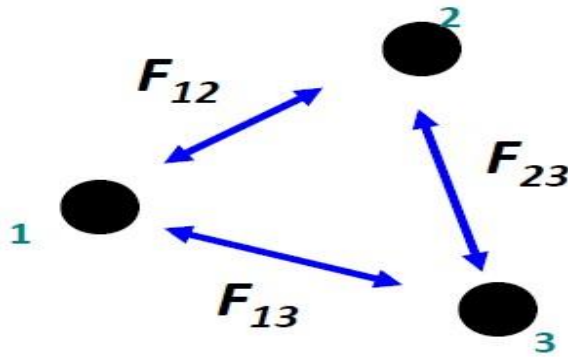
Summing the two equations,

$$r(t + \Delta t) \approx 2r(t) - r(t - \Delta t) + \frac{f(t)}{m}\Delta t^2$$

Now we have advanced our atoms to time $t + \Delta t$!!

Velocity of the atoms:
$$v(t) = \frac{r(t + \Delta t) - r(t - \Delta t)}{2\Delta t}$$

Three Atoms



$$F_{ij} = -\nabla U_{ij}$$

$$F_i = \sum_{j \neq i} F_{ij}$$

$$F_1 = F_{12} + F_{13}$$

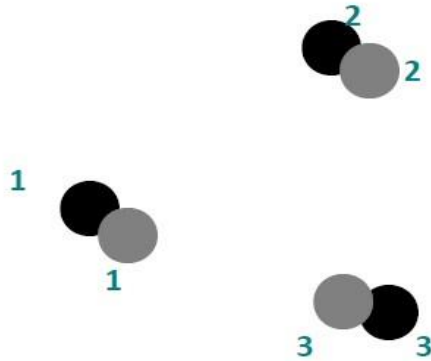
$$F_2 = F_{21} + F_{23}$$

$$F_3 = F_{31} + F_{32}$$

Newton's IInd Law:

$$a_i = F_i / m_i$$

Progress in time



$$\underline{r(t) \rightarrow r(t + \Delta t)}$$

$$x(t) \rightarrow x(t + \Delta t)$$

$$y(t) \rightarrow y(t + \Delta t)$$

$$z(t) \rightarrow z(t + \Delta t)$$

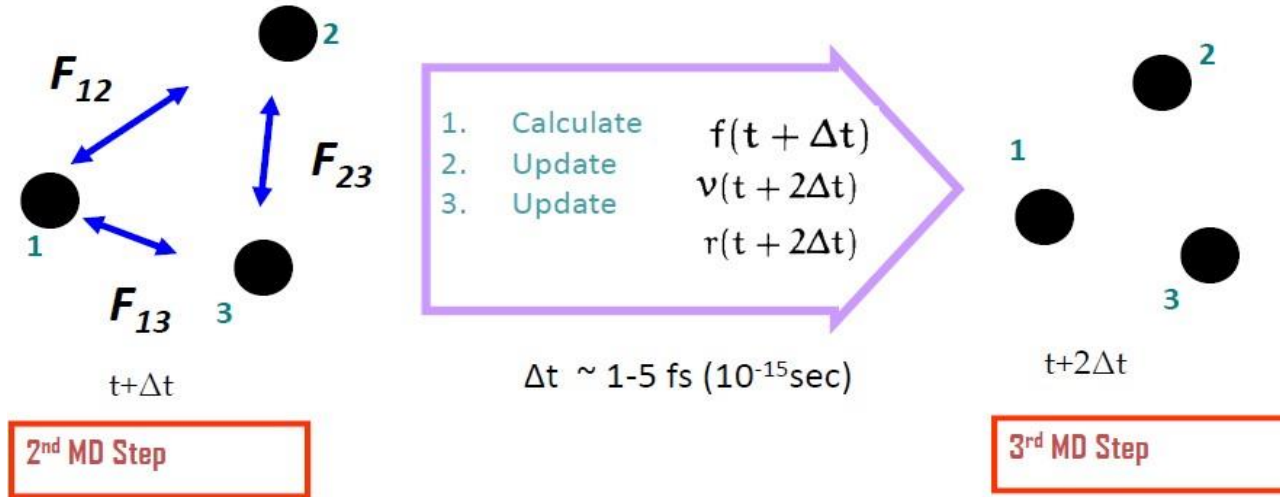
$$\Delta t \sim 1-5 \text{ fs } (10^{-15} \text{ sec})$$

Advance positions & velocities of each atom:

$$r(t + \Delta t) \approx 2r(t) - r(t - \Delta t) + \frac{f(t)}{m} \Delta t^2$$

Velocity of the atoms: $v(t) = \frac{r(t + \Delta t) - r(t - \Delta t)}{2\Delta t}$

Atoms move forward in time

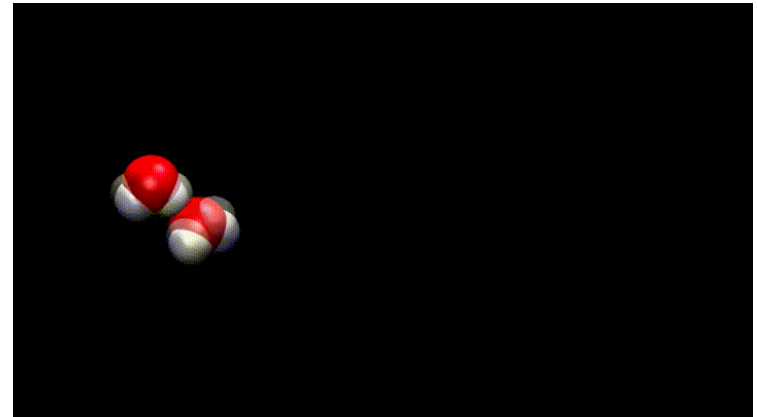
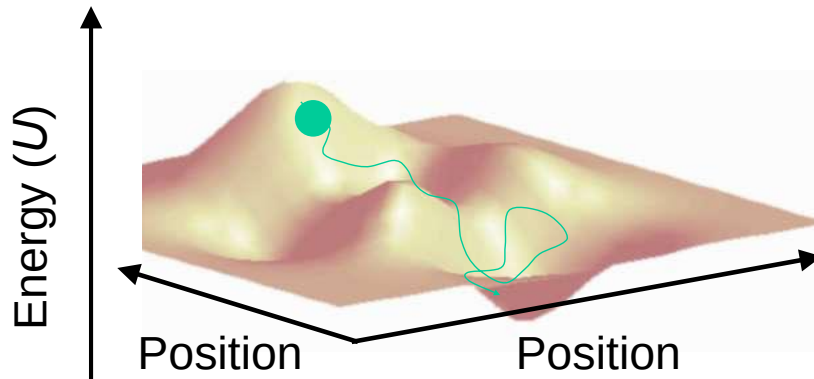


Continue this procedure for several lakhs of Steps. Or as much as you can afford!

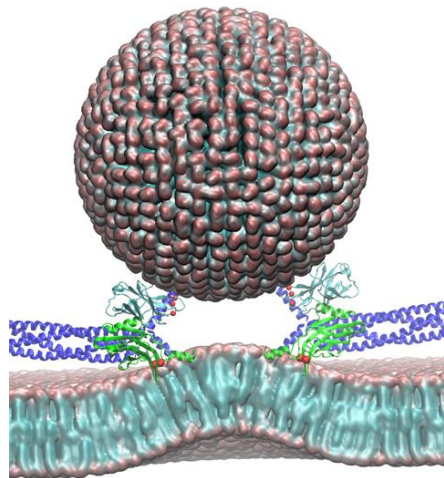
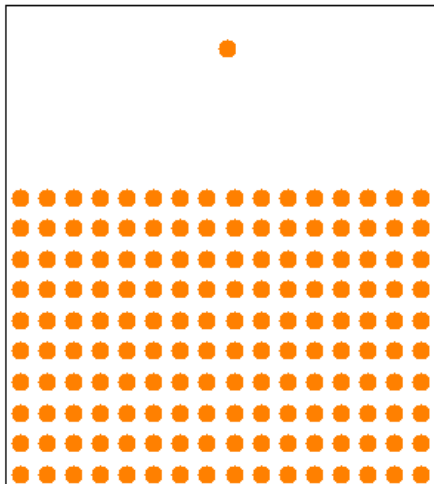
The main O/P of MD is the trajectory.

Molecular Dynamics Simulations

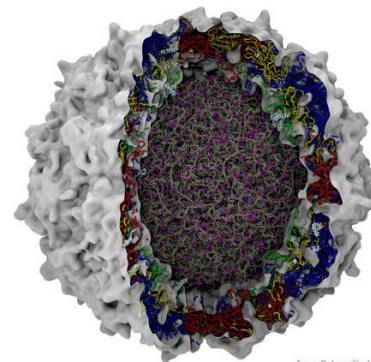
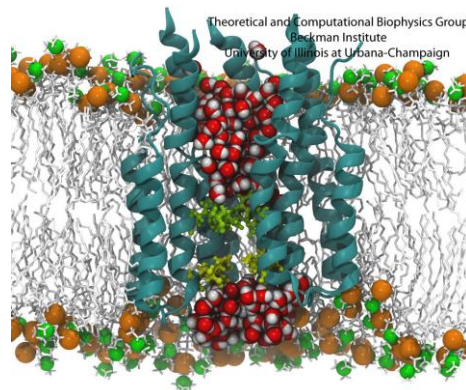
- Build realistic atomistic model of the system
- Simulate the behavior of your system over time using specific conditions (temperature, pressure, volume, etc.)
- MD simulations model the motion of some group of particles by solving the **classical equations of motion**.



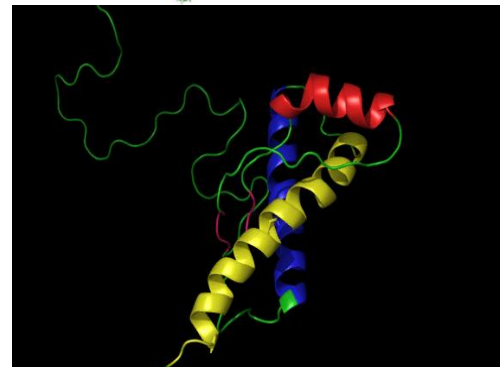
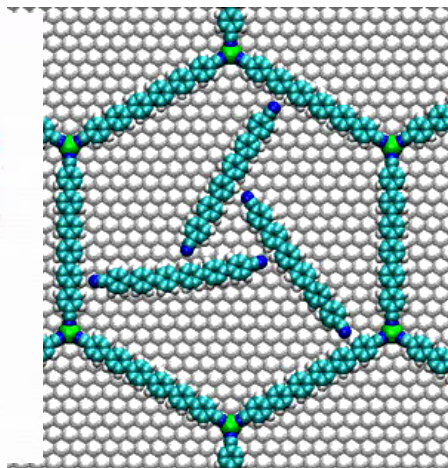
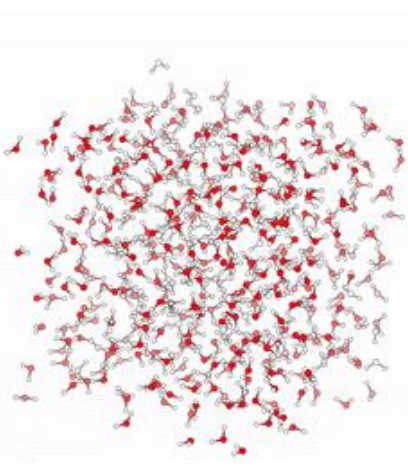
time 0.0041 ps



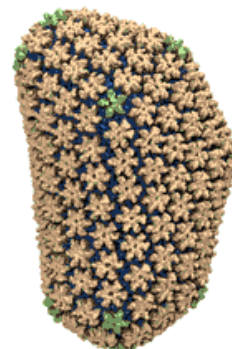
MD Systems



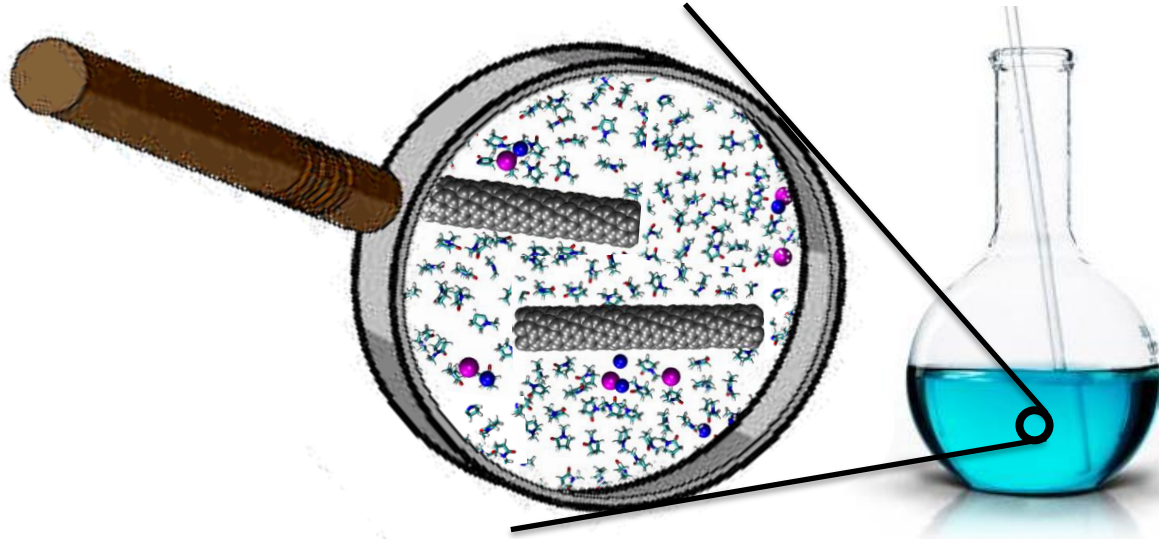
Jason.Roberts@nsl.org.au
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0.2 ns

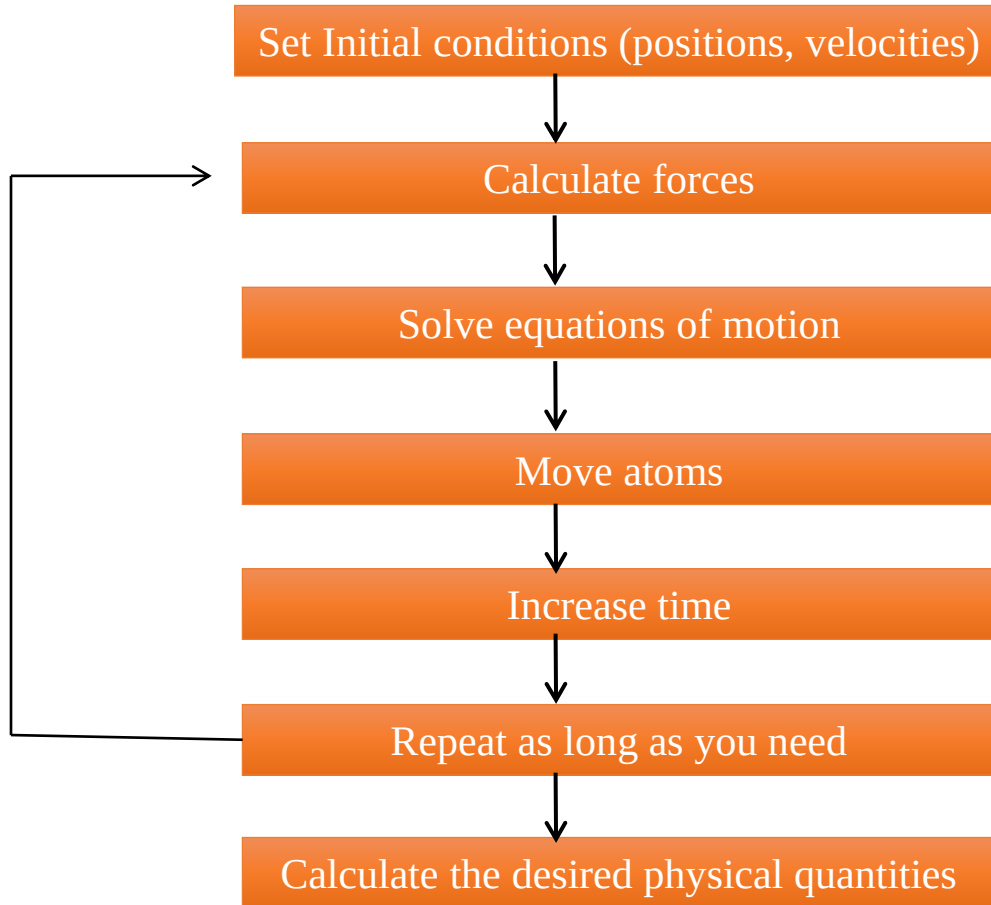


Molecular Dynamics



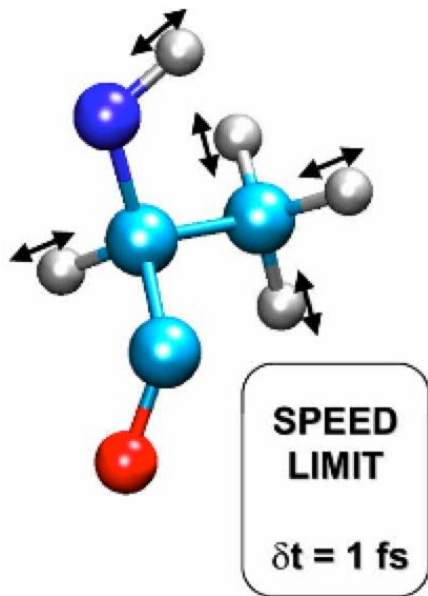
MD is a computational microscope. Understanding how things work on a detailed *molecular* level ranging from electronic structures to long-time phase behavior of molecules

MD algorithm



Time Step δt

- Time step δt should be as large as possible to still get accurate trajectories (on the time scale needed) and conserve of energy.
- In general, δt should be ≈ 0.01 x the fastest behavior of your system

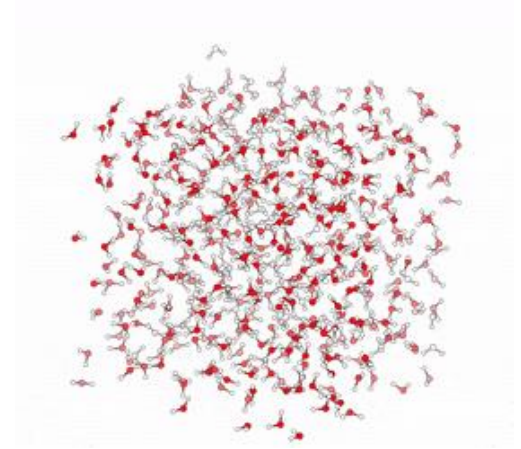


Choosing the timestep: for numerical stability and accuracy in conserving the energy, one typically needs to **pick a time step that is at least one order of magnitude lower than the fastest time scale in the system.**

Atoms oscillate about once every 10^{-14} s in a solid $\Rightarrow$
MD time steps are $\approx 10^{-15}$ s in simulations of solids.

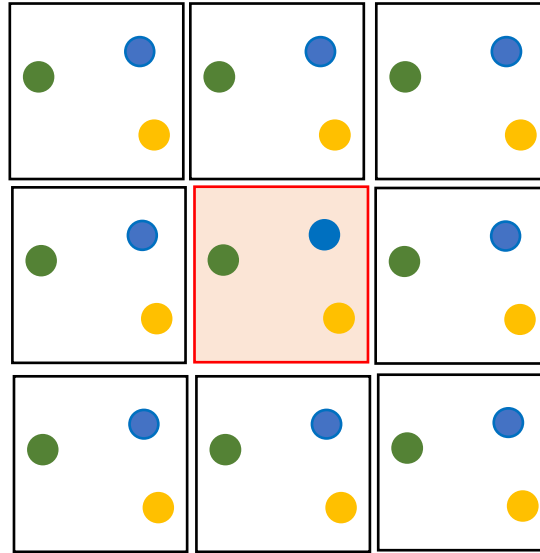
Geometry and boundary conditions

- Most simulations probe the structural and thermodynamic properties of a system of a few thousand particles.
- For such small systems, the choice of boundary conditions can strongly affect the measured properties.
- If nothing special is done, atoms near the boundary would have less neighbours than atoms inside.
- This causes surface effects in the simulation to be much more important than they are in the real system.



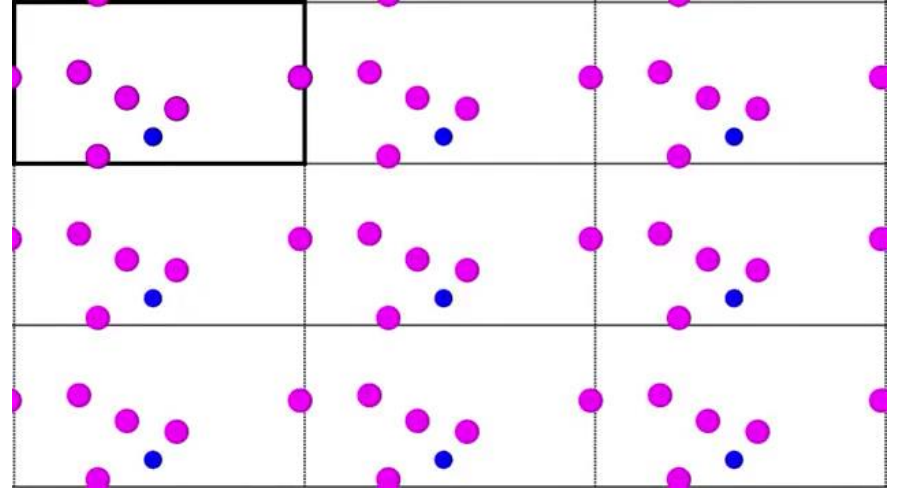
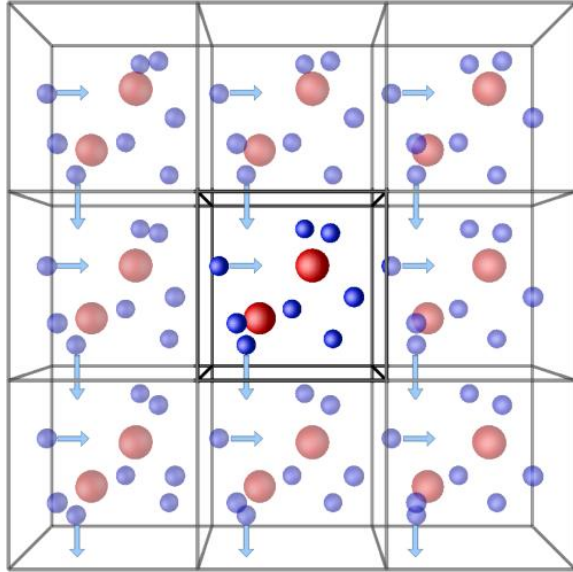
Periodic boundary conditions

Consider box of size L ,
repeat box infinitely many
times in all directions



Basic box

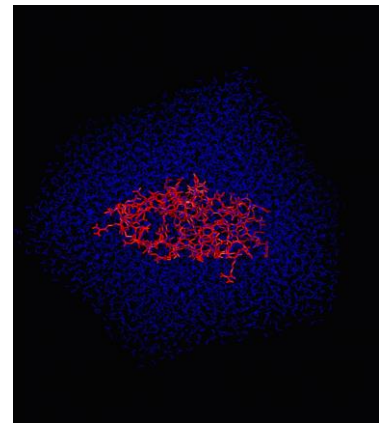
copies



a particle which goes out from the simulation box by one side is reintroduced in the box by the opposite side.

Solvation

- Biological activity is the result of interactions between molecules and occurs at the interfaces between molecules (protein-protein, protein-DNA, protein-solvent, DNA-solvent, etc).
- Many biological processes occur in aqueous solution
- Solvation effects play a crucial role in determining molecular conformation, electronic properties, binding energies, etc
- How to model solvation?
- **explicit treatment**: solvent molecules are added to the molecular system
- **implicit treatment**: solvent is modeled as a continuum dielectric



MD – Shortcomings

- Quality of the forcefield
- Size and Time – atomistic simulations can be performed only for systems of a few tenths of angstroms on the length scale and for a few nanoseconds on the time scale
- Conformational freedom of the molecule – the number of possible conformations a molecule can adopt is enormous, growing exponentially with the number of rotatable bonds.
- Only applicable to systems that have been parameterized
- Connectivity of atoms cannot change during dynamics – no chemical reactions.

MD

Programs:

AMBER

CHARMM

VASP (DFT)

XMD

CPMD(DFT) (Car-Parrinello MD)

NAMD

Web-servers for running MD Simulations

- <http://biocomp.chem.uw.edu.pl/CABSflex2/submit>
- <http://vienna-ptm.univie.ac.at/>
- <https://www.playmolecule.org/SimpleRun/>
- <http://chemyang.ccnu.edu.cn/ccb/server/LARMD/>